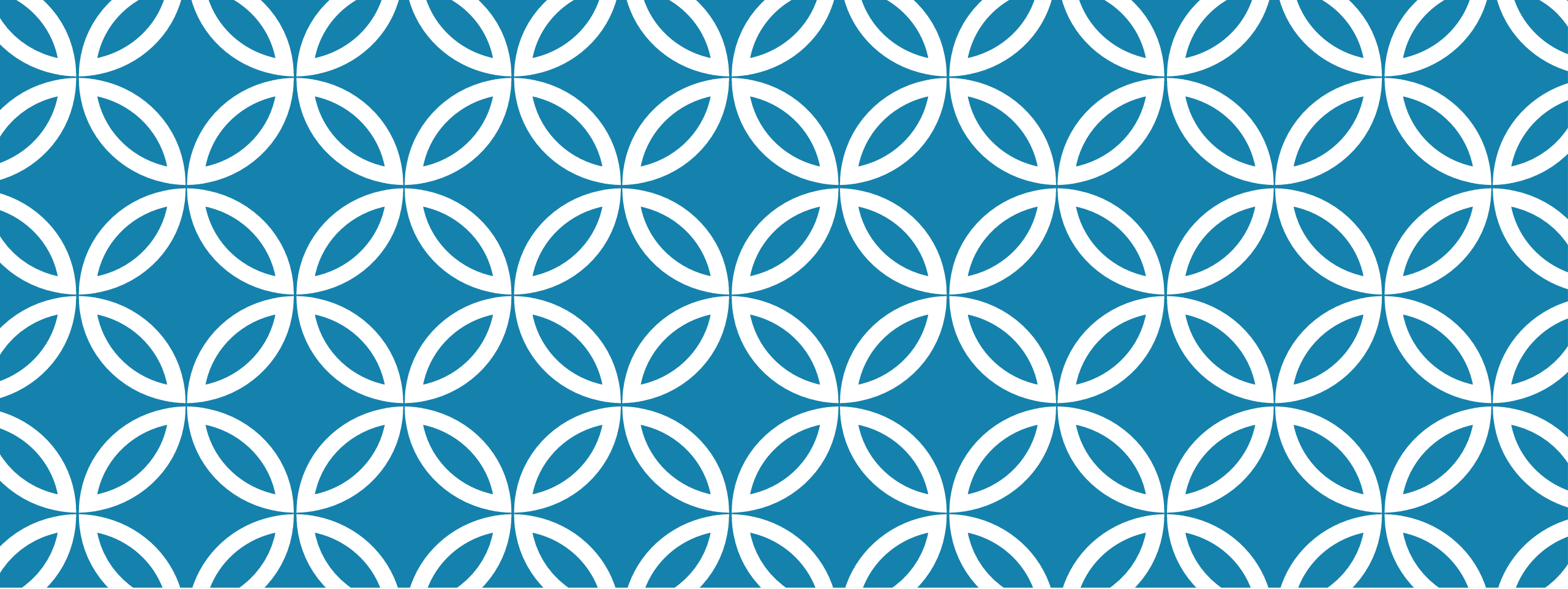


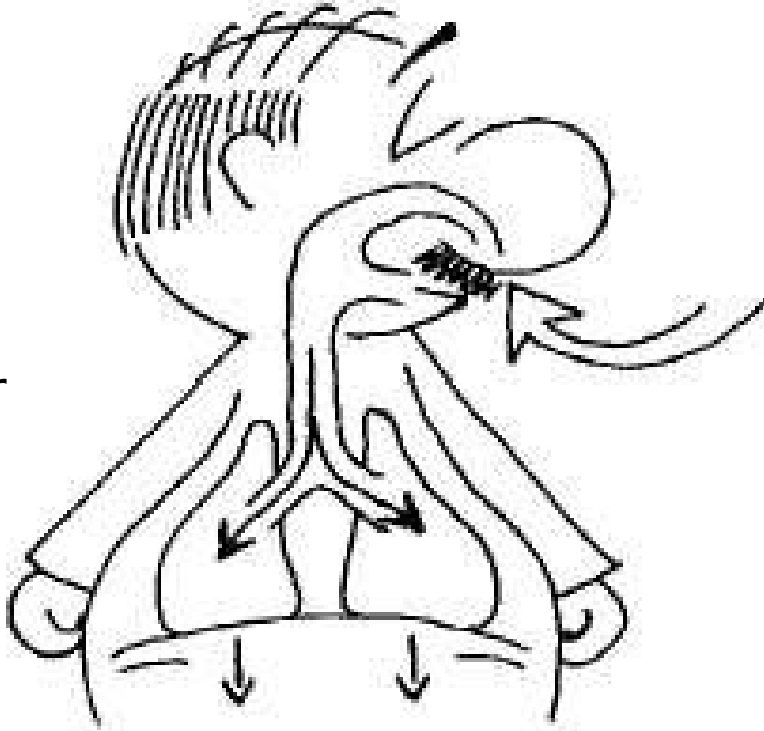


BIO262 **PHYSIOLOGY-**

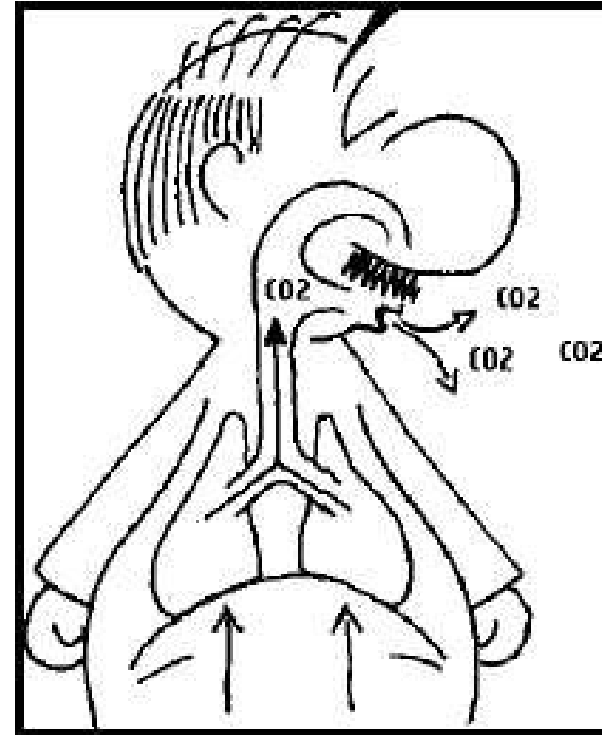
**Module 12 -
Respiratory
System**

VENTILATION- BOYLES LAW

Inhale:
Diaphragm contracts and drops making the thoracic cavity bigger. Intrapulmonary pressure **DROPS** below atmospheric: air moves in.



Inhalation



Exhalation

Exhale:
Diaphragm relaxes and returns making the thoracic cavity smaller. Intrapulmonary pressure **INCREASES** above atmospheric: air moves out.

QUESTION

Which statement is true about breathing?

A- When the diaphragm relaxes, the thoracic cavity increases in size and air moves into the lungs.

B - When I hike in the mountains the pressure differences are greater, so I move more air into my lungs.

C- The diaphragm contracts and drops. This increases the thoracic cavity size and air moves into the lungs.

D - Air flows from an area of less pressure to an area of greater pressure.

RESPIRATION

External

Moving air
into and out
of lungs

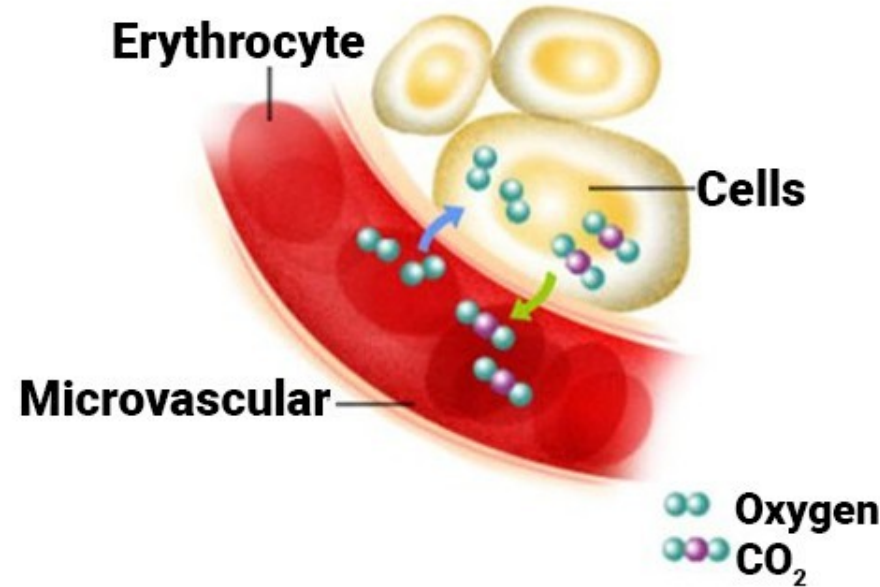
Uses
diaphragm
and external
intercostal
muscles –
both contract!



**Between environment
and lungs**

Internal

Exchange of gasses
- Between blood and cells



Between blood and cell

Oxygen is
bound to
hemoglobin in
the red blood
cell.
Blood flowing to
tissues =
perfusion

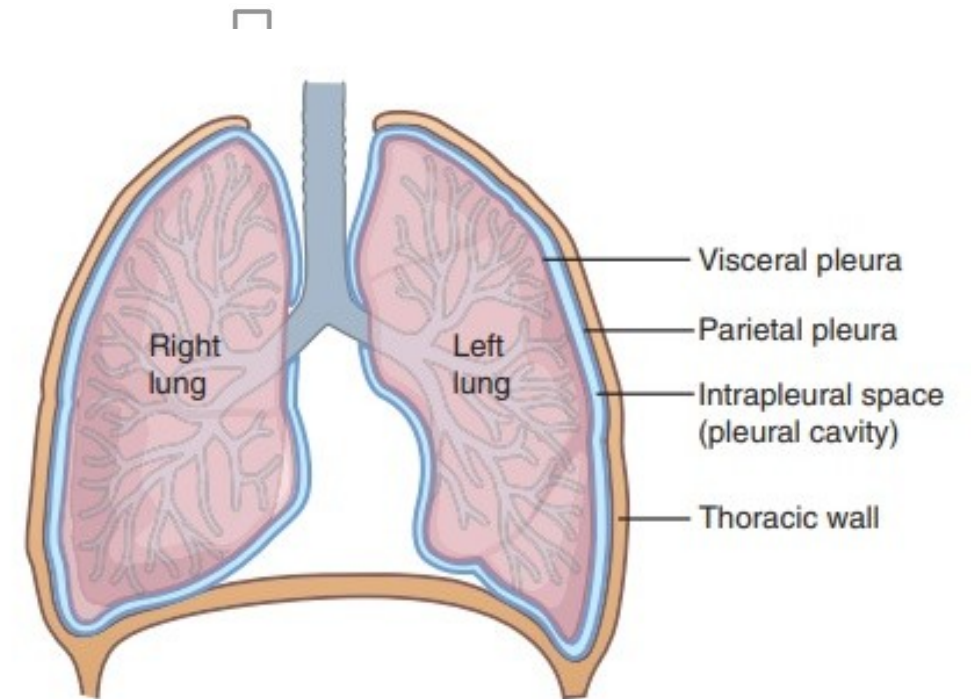
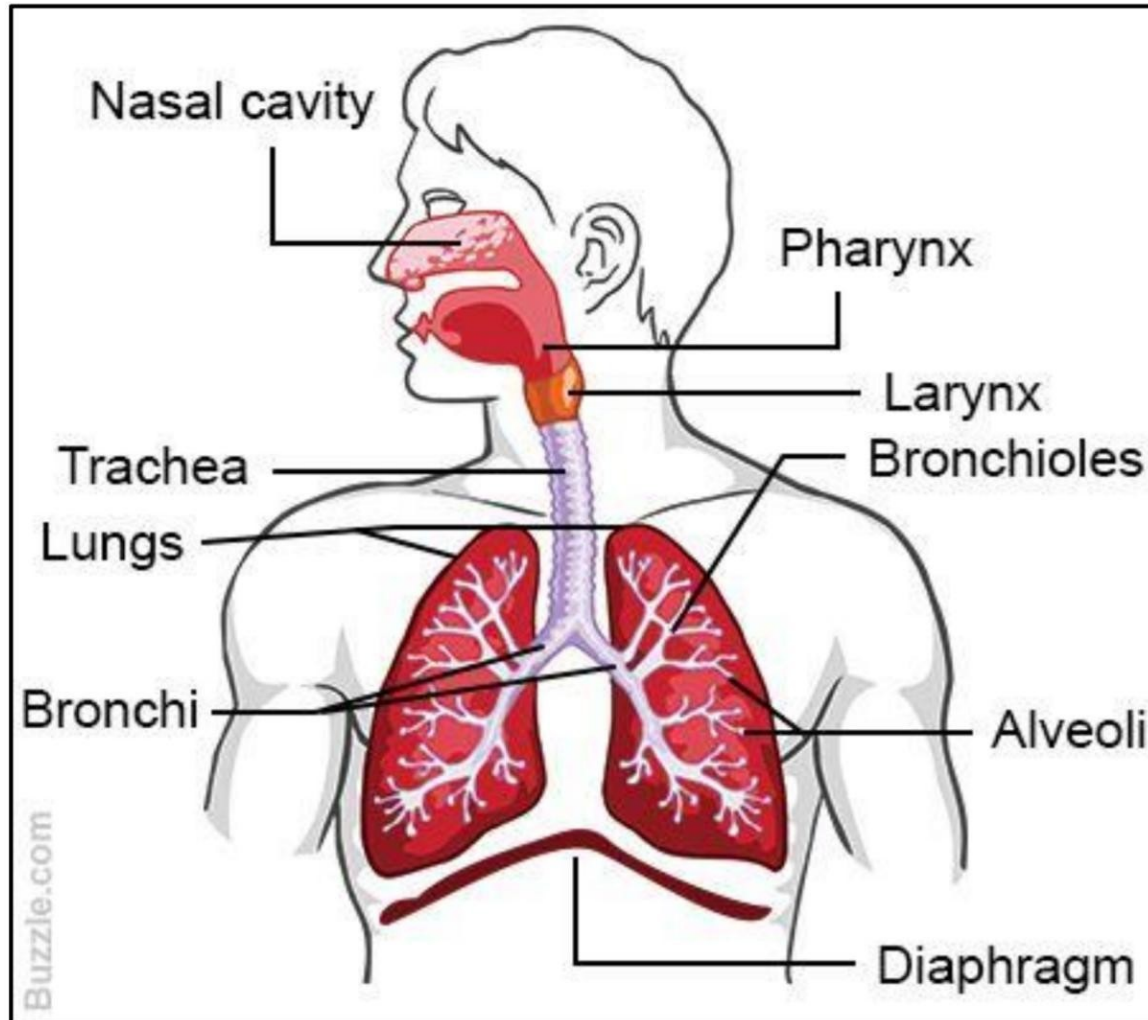
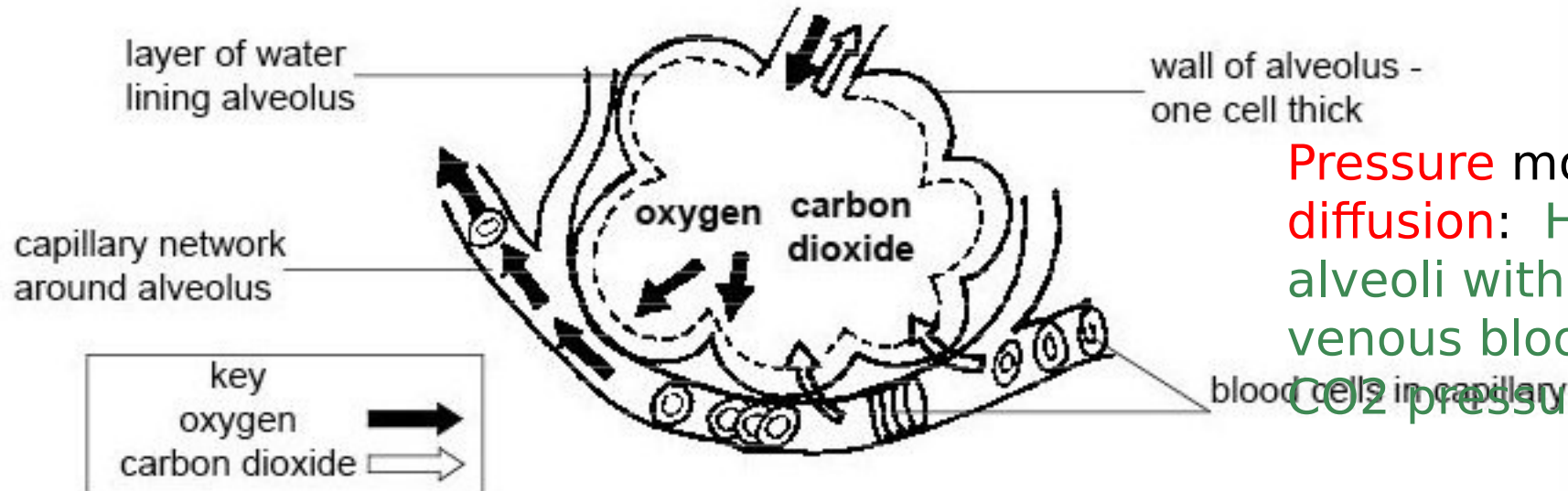
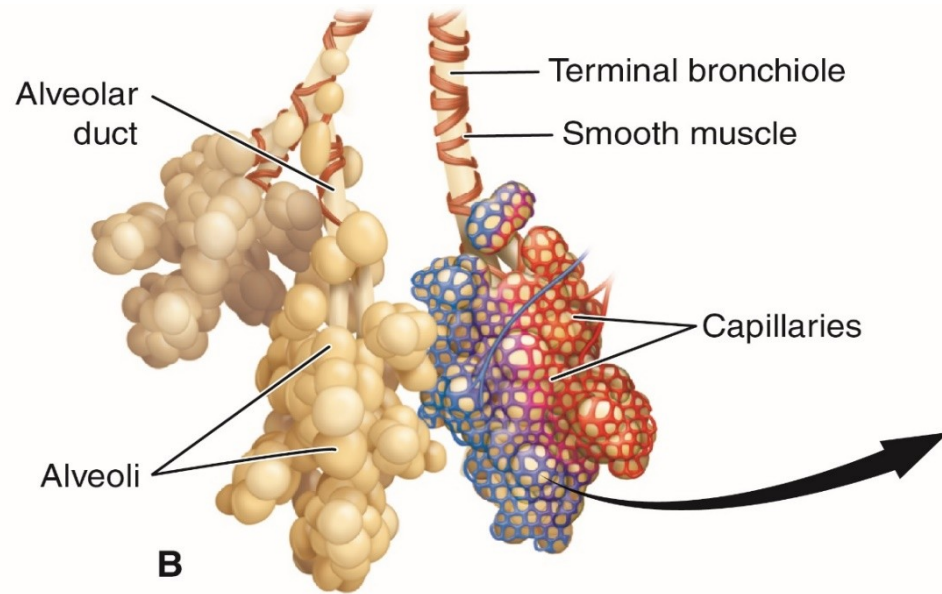


FIGURE 21-7 The lungs reside in the pleural cavities, subdivisions of the thoracic cavity. They are lined with a serous membrane called the pleura. The intrapleural space is located between the visceral and parietal pleura.

Alveoli -

- Functional unit (gas exchange)
- Simple squamous epithelium

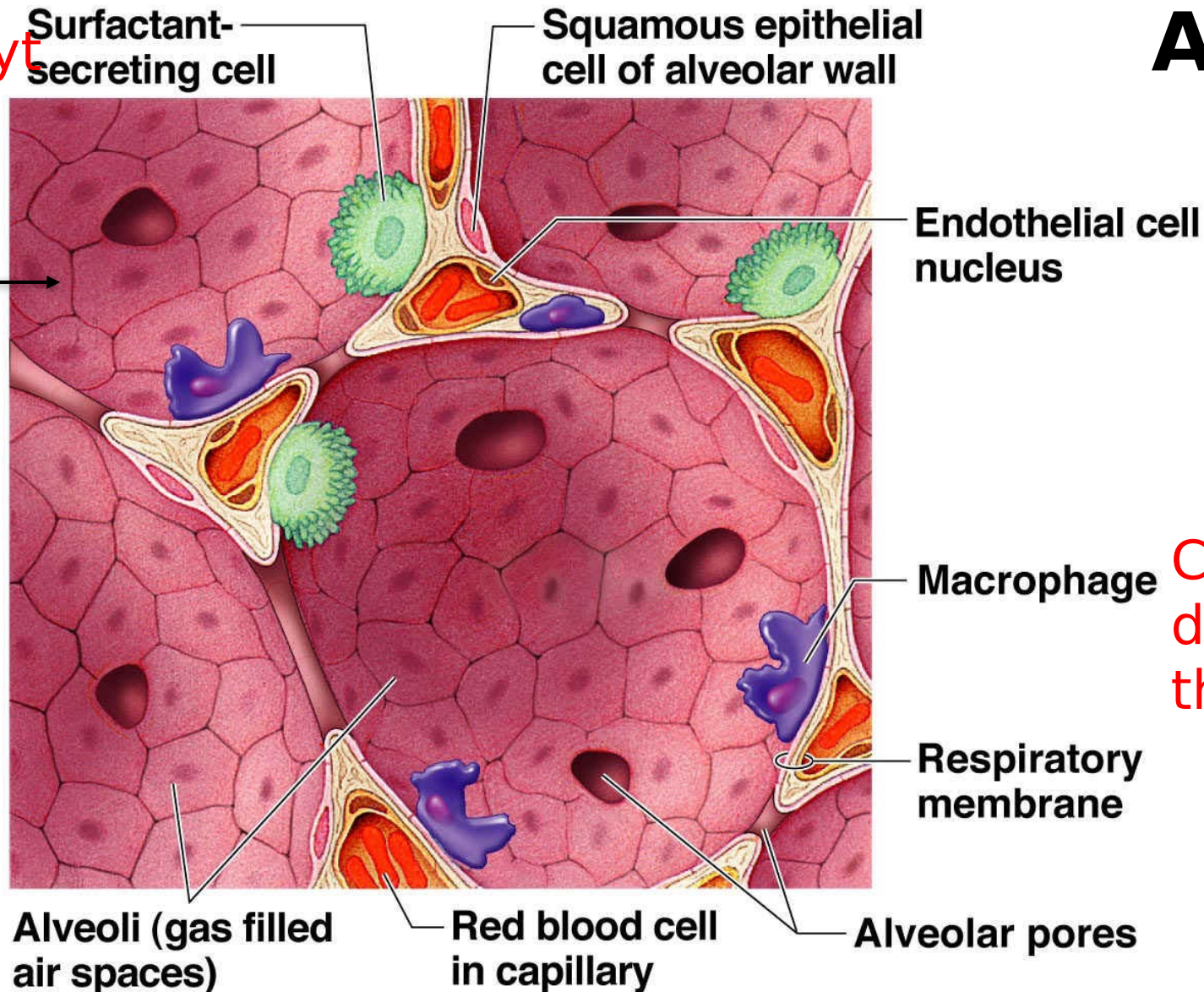


Pressure moves gasses via diffusion: High O₂ pressure in alveoli with low O₂ pressure in venous blood from heart. High CO₂ pressure in this blood.

Alveoli -

Type II
Pneumocyte
e=

Type I
Pneumocyte
=
Gas
exchange!
Simple
squamous!

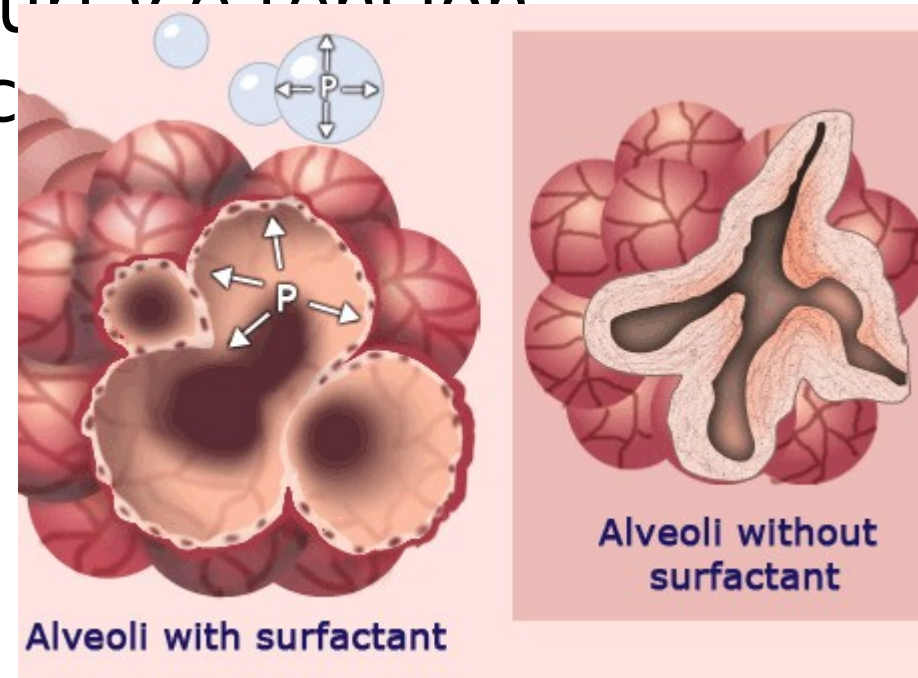
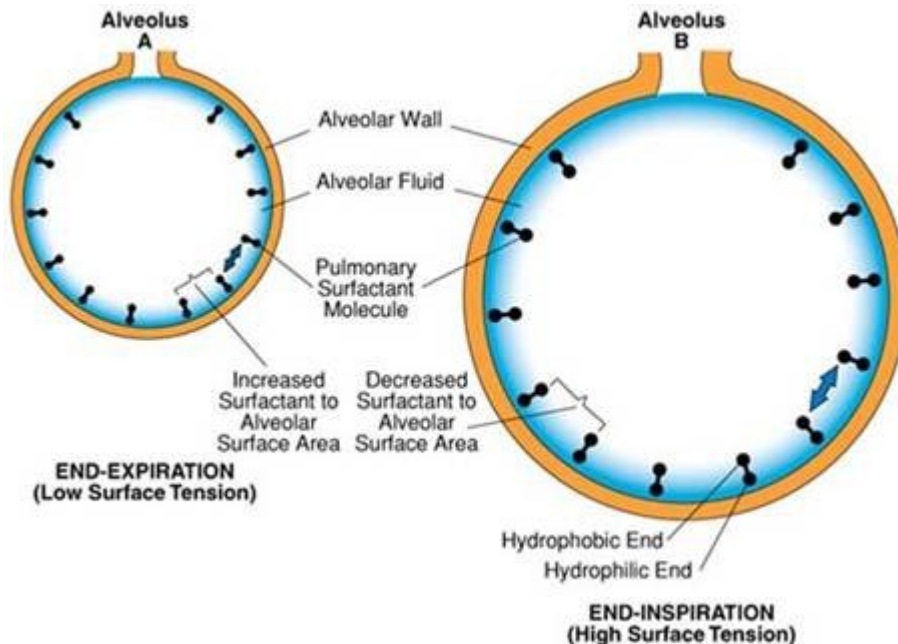


Cleans up any
debris - foreign
things

WORK OF BREATHING

Lung Compliance:

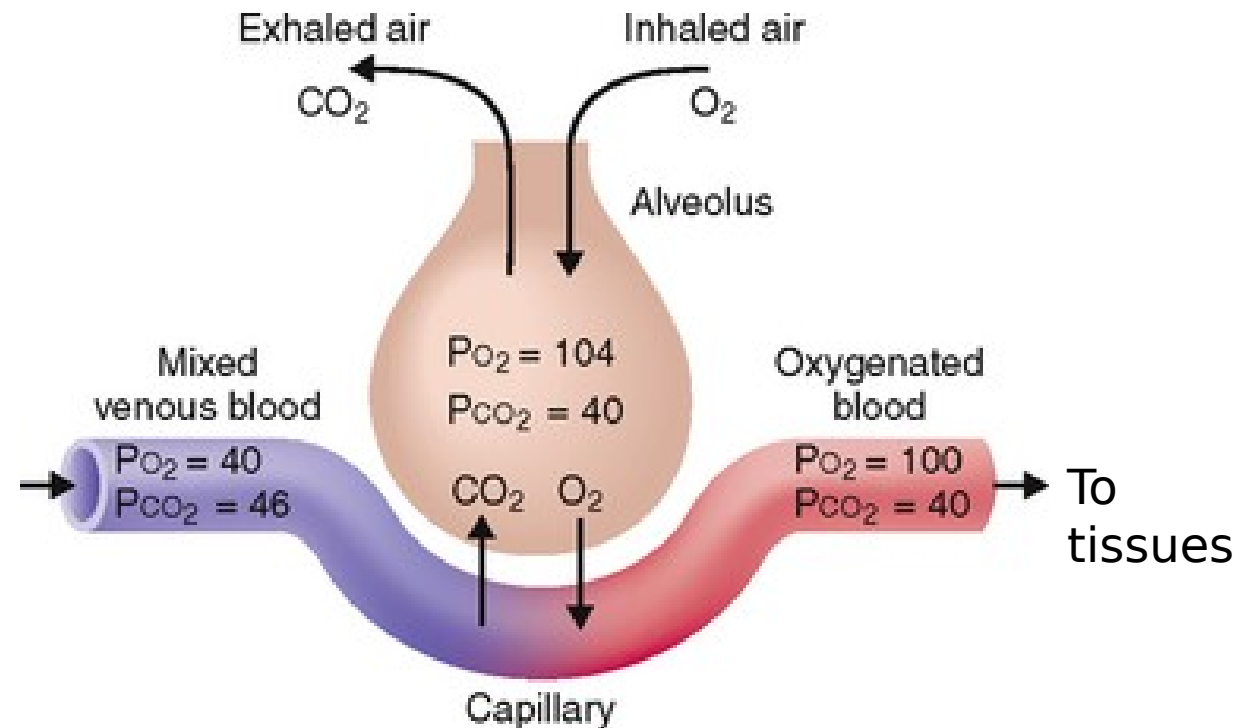
- Ability of lung tissue and chest wall to expand
- **Surfactant** – Prevents alveoli collapse by decreasing surface tension



MOVEMENT OF GASSES

Enhanced by:

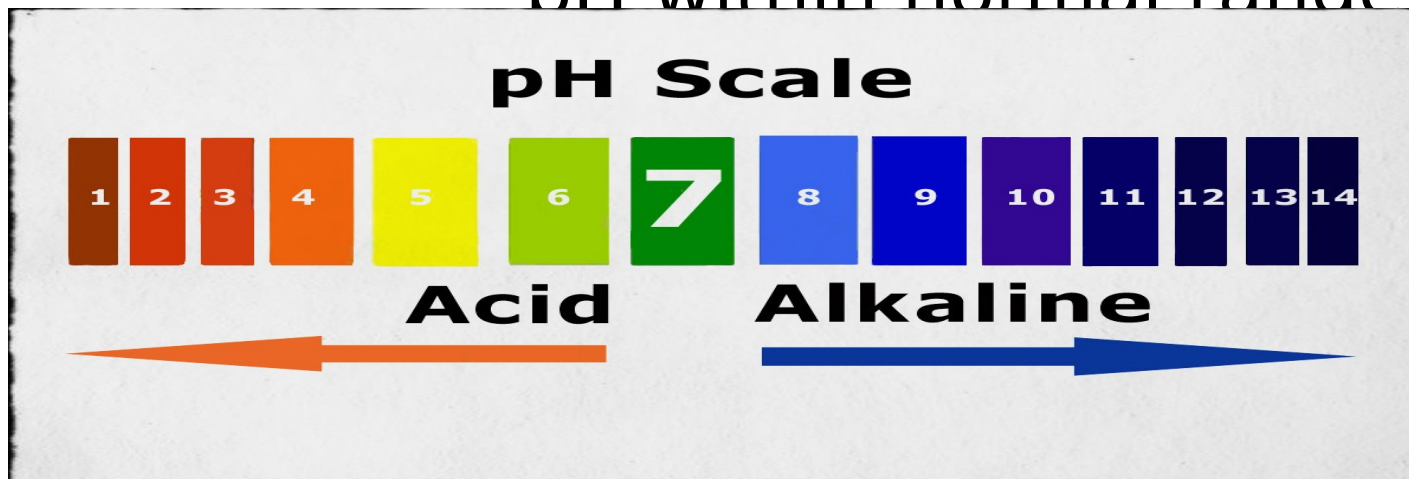
- **Pressure** – concentration gradient
- Solubility of gases: CO₂ more soluble = diffuses more easily
- Distance from alveoli to capillary = super short
from tissues
very thin membranes
- **Large surface area**



ACID/BASE -

- **CO₂ is an acid**
 - In serum/blood = **carbonic acid**
 H_2CO_3
 - In Lungs = H_2O and CO_2
- Kidneys produce **bicarbonate** which acts as a **buffer** in the blood and keeps blood pH within normal range

Accumulation of CO_2 in blood = acidosis



Normal blood levels = 7.35 – 7.45

acidosis = anything below 7.35

alkalosis = anything above 7.45

QUESTION -

Edward had really bad heartburn, so he mixed up some baking soda with water and drank it. Because of the amount of baking soda, Edward now has serum (blood) pH of 7.48. What type of breathing would you expect to see in Edward?

- A – Hyperventilation
- B – No change in rate or rhythm
- C – Hypoventilation
- D – He would stop breathing

QUESTION

Charles is a premature newborn. He was born at 32 weeks. He is struggling to breath. Why?

A – His lungs are collapsed due to the trauma of birth.

B – He is not yet producing surfactant, so his alveoli are difficult to inflate.

C – Because he is premature his bronchi are not yet open.

D – His diaphragm muscle isn't working yet.

REMINDER



DUE WEDNESDAY: 1st
discussion post

Pre-lesson checkpoint

Module 12 quiz
(Respondus)

Peer response in discussion
Lab quiz (Respondus)

EXAM 5 opens MONDAY –
covers modules 10
through 12

You can avoid feeling overwhelmed on Sunday by completing something each day – this also leads to better retention of knowledge.



DUE SUNDAY:

Struggling?
Let's meet – send me an email and we can arrange a meeting time!